Prof. Tashfeen
CS 2163: Java

Assignment 2

Question 1. Run the command `java --version`, give the output. What does it mean?

Question 2. What are the commands `cd` and `ls` (`dir` if you use Windows) for? If you just opened your console (terminal/command prompt/power shell), how can you change the *current working directory* to desktop? Give the full command. If you now run `ls` or `dir`, does this look familiar? What does the output mean? Use an online search engine for this question if you don't know the answer.

Question 3. Use the Java Development Kit Version 19 API Specification to find the documentation for the `java.lang.String.charAt(int)`. What is the return type of this public function of the String class? What does it do?

Question 4. What does `"Bonjour!".charAt(0)` return?

Question 5. Watch *Sigma notation for sums*, write the following in the sigma notation,

$$2^1 + 2^2 + 2^3 + \cdots + 2^{10}.$$

Question 6. Read the relevant parts of the textbook as described in the online classroom. Write a Java program that reads some positive integer n and prints out a table with the Gregory series approximations π_i and their distance (error) from the real π, i. e., $|\pi_i - \pi|$.

The Gregory series is given as,

$$\pi_n = \sum_{i=1}^{i=n} \frac{4(-1)^{i+1}}{2i-1} = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \cdots + \frac{4(-1)^{n+1}}{2n-1}$$

You may start by modifying the program in code listing 1 that prints out "Hello World!" to the console.

```java
/**
 * Name: Tashfeen, Ahmad
 * File: TashfeenAhmad.java
 * Description: Program to print out "Hello World!"
 */
public class TashfeenAhmad {
  public static void main(String[] args) {
    System.out.println("Hello World!");
  }
}
```

Listing 1. A complete Java program to print "Hello World!"

Save the listing 1 code in a file called `TashfeenAhmad.java` (use your own name, see requirement 4) and run,

```
javac TashfeenAhmad.java
java TashfeenAhmad
```

6.1. Requirements

Note that *these are not merely suggestions*. You will lose points (possibly all of them) if you don't fulfil these. Even if your program works.

1) If the user inputs $n < 1$, your program should print, "Warning: n is a positive integer." and finish executing gracefully.

2) Your output should be well formatted in a table with three aligned columns. They should contain the value of the iteration variable $i \leq n$, the current approximation of π at the ith iteration π_i and the error $|\pi_i - \pi|$.
3) Your table must include a header, e. g., `| i| Pi'| Error|`.
4) Your class file should be named `LastnameFirstname.java`, e. g., for me it is `TashfeenAhmad.java`. If you used the listing 1 code then don't forget to change the class name (and therefore the filename).
5) You must close your scanner.
6) Your program must compile and run from the command line using `javac` and `java` commands.
7) If you've been asked to demonstrate your program to the professor then you'll need to do so in his office hours or make an appointment. If you've been asked to demo your program and you fail to do so, you will receive a zero in the assignment.

6.2. Hints

You'll need to import the scanner class and read in an integer n. Make sure to close the scanner once you're done using it. You'll need a for-loop but unlike most for-loops, your iterating variable i will need to start at 1 and run till $i \leq n$ instead of strictly less than n. You can use `Math.PI` to get the Java double point precision approximation of the true π. For the purposes of this assignment, we'll consider this value equal to the true π. Similarly, you may use `Math.abs` and `Math.pow` to compute the absolute values and powers

For table formatting, do you see how you can use the following print statements?

```
System.out.printf("|%3s|%10s|%10s|%n", "i", "Pi'", "Error");
System.out.printf("|%3d|%10.7f|%10.7f|%n", i, pi, error);
```

What variables do you need to define?

6.3. Example Executions

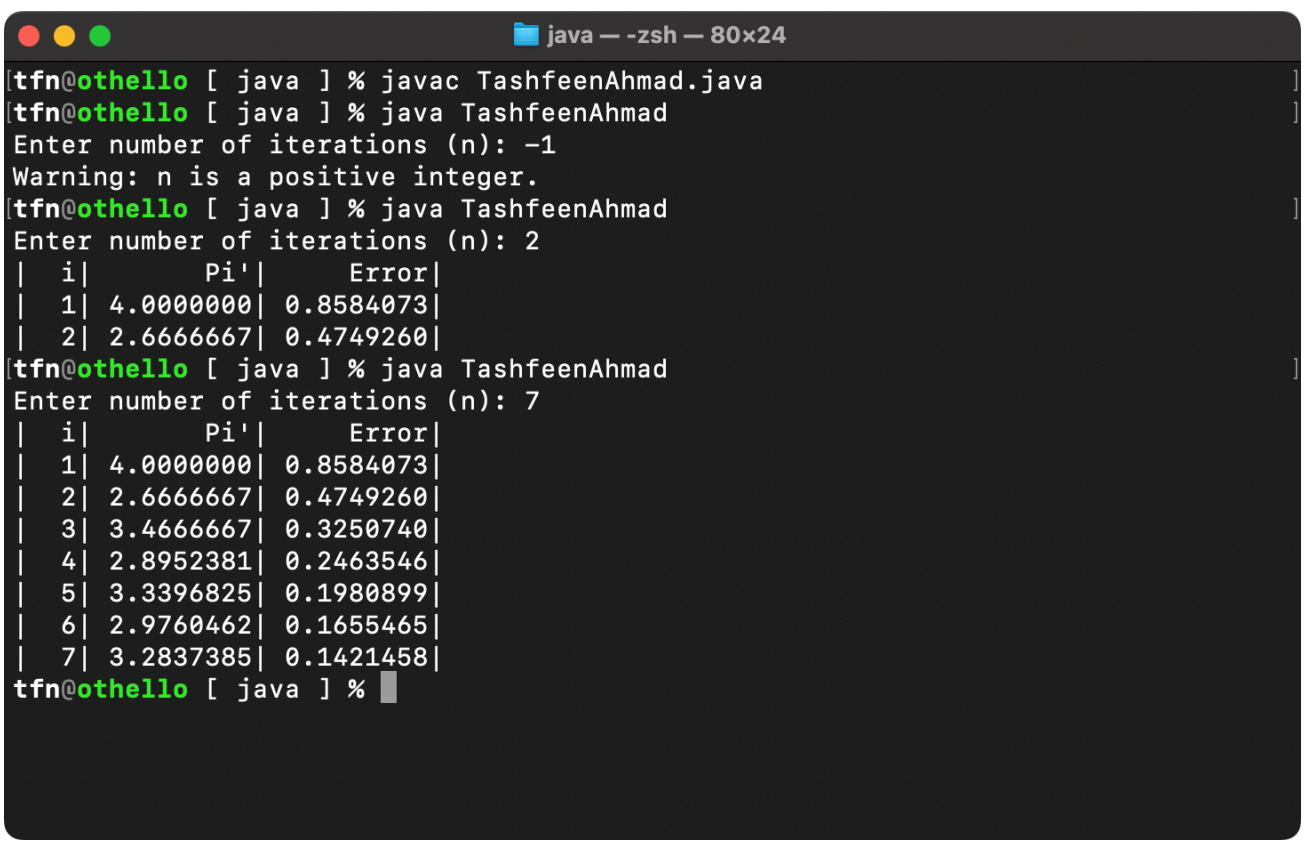

Figure 1. Example execution for the assignment 2 program.

Question 7. What was the hardest part of this assignment for you? Why do you think you struggled with this?

Submission Instructions

1) A PDF document containing the answers to all the questions. For question 6, include a a screenshot like the one in figure 1.
2) Submit your `LastnameFirstname.java` file. Note that this is not the `LastnameFirstname.class` file.

You will not receive any credit if any of the files to be submitted are missing.